

Health Data Mapping, Insight Discovery and AI Readiness

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Data Quality Assessment

Strengths:

- 1. Data Versatility** By successfully cataloging APIs across the 4 main brands (Apple, Garmin, Google, and Samsung), the data system does show high versatility. The clinic or platform does not need to force patients to buy a specific brand watch. It can meet patients exactly where they are, utilizing the devices they already wear.
- 2. Standardisation** The creation of the Data Dictionary and the Ikarious Standard Variable (e.g. creating a single *body_fat_percentage* column to catch both Apple's *bodyFatPercentage* and Google's *Body Fat*). This helps with mapping these varying inputs to a unified, standardised schema, the system achieves semantic interoperability. It ensures that when the AI or the doctor queries the database for "Sleep duration", they instantly get the combined data from all Apple, Garmin, Samsung, and Google users.
- 3. Comprehensive Biometric Depth** The Data Source Catalogue proves that the system is capturing clinical-grade physiological metrics. The platform captures a holistic view of objective physical health, spanning Cardiovascular (Blood Pressure, ECG), Mobility (Gait asymmetry, step length), and Clinical Markers (Blood Glucose, Lean Body Mass). Through integrations like the Garmin Health API, the system can capture deep, high-resolution data.
- 4. AI and Machine Learning Readiness** Because the Data Dictionary explicitly defines the Data Type (e.g. Numeric, Boolean) and the Unit (e.g., mg/dL, Minutes) for every standard variable, the database is perfectly pre-structured for predictive modelling. The engineering team has exactly what they need to build the ETL (Extract, Transform, Load) pipelines to feed the AI.

Weaknesses:

- 1. The Psychosocial Gap:** The wearable ecosystem is blind to social determinants of health. There is a gap in data regarding patient relationships, spirituality and mental health. This will need designing questionnaires and screening tools to measure such determinants.
- 2. Quantitative Dietary Data:** Current Google and Samsung integrations offer "Nutrition Logs" focused on quantitative calorie and macronutrient counting, which is inaccurate and misaligned with longevity science that focuses on the quality of the diet.
- 3. Consistency:** Different brands measure the same concept using different data structures. For example, Apple records Blood Oxygen as a decimal percentage (*HKQuantityTypeIdentifier.oxygenSaturation*), while Samsung records it as a whole integer via Google Health Connect (*OxygenSaturationRecord*). Without a strict backend translation layer, the AI will misinterpret these structural differences.

Data Gap Assessment

Mapping the hardware data against the core pillars of longevity science reveals four main data gaps. To calculate a true, holistic iKfactor, the following data gaps must be acknowledged and addressed:

Gap 1: The Psychosocial Data Gap Negative relationships, lack of life purpose, and clinical depression are primary drivers of biological decline. The wearable ecosystem currently remains blind to these critical social determinants of health.

Gap 2: The Qualitative Nutrition Gap Current integrations offer nutrition logs heavily focused on quantitative calorie and macronutrient counting, which is inaccurate and misaligned with longevity science that focuses on the qualitative aspects of the diet.

Gap 3: The Behavioral Risk Gap The platform currently lacks the baseline data required to stratify a user's behavioral mortality risk (such as smoking or binge drinking patterns).

Gap 4: The Normalisation Gap A gap remains in the execution of this data. Because Garmin uses math to calculate a "Stress Level" and Apple uses raw "HRV," the platform still lacks the ETL (Extract, Transform, Load) normalisation engine required to mathematically harmonize these different hardware algorithms. Doctors cannot compare a Garmin user's baseline directly against an Apple user's baseline.

Signal vs. Noise Analysis

1. Context-Free Stress Scores

- **The Noise:** Wearables use HRV and heart rate to output a "Stress Score". A Garmin watch might say "Stress Score of 60" while Apple outputs an "HRV of 40ms." This is purely algorithmic noise driven by the different math of the tech companies.
- **The Signal:** The AI must evaluate patients against their own baseline (Z-Scoring), not against each other. The true clinical signal is extracted by tracking the "Rate of Change". A valid signal triggers only when Patient A's HRV drops 2 standard deviations below their own 30-day Apple baseline, or when Patient B's stress score spikes above their own Garmin baseline. This turns fragmented relative data into an accurate, comparable clinical alert.

2. Quality of Movement vs. Quantity of Steps

- **The Noise:** Daily Step Count. A patient logging 2,000 steps instead of 10,000 steps on a one day is usually just scheduling noise (e.g. bad weather) and does not indicate a sudden decline in biological health.
- **The Signal:** A longitudinal, three-month increase in *walkingAsymmetryPercentage* paired with a decrease in *walkingSpeed* (captured via Apple Health). Because this is happening steadily over 90 days, the AI knows it is not just a temporary situation, it is an established health change.

3. Sleep Interruptions vs. Respiratory Events

- **The Noise:** Brief tossing and turning as part of normal sleep architecture or minor awakenings at night may be detected and registered as sleep disturbance noise.
- **The Signal:** A sharp drop in Oxygen Saturation occurring at the exact same timestamp that the Respiratory Rate spikes. This isolates a critical longevity risk: Obstructive Sleep Apnea (OSA) from its noise.

4. Subjective Signals

- **The Noise:** A spike in Apple’s *electrodermalActivity* or Garmin’s Stress Level could be acute sympathetic nervous system activation from a hard workout or a busy day at work. The hardware alone cannot tell if the stress is physical or emotional.
 - **The Signal:** Anchoring the wearable data against standardised qualitative tools such as PROMIS Social Isolation or Global Mental. If a chronically elevated wearable stress baseline intersects with a PROMIS Social Isolation or Global Mental Health T-Score above a cut-off value, the AI creates a high-value signal as: “Chronic Stress”. The clinician immediately knows the physiological damage is being driven by psychological factors rather than physical strain.
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Recommendations for Future Data Collection

A. For the Objective Data:

The data engineering team must build a Data Normalization Engine referred to as an ETL pipeline: Extract, Transform, Load. This engine acts as a universal translator between the raw wearable APIs and the physician’s dashboard. Here is the step-by-step technical strategy to do so:

1. Define the Data Schema Before translating, we must define exactly what the final output should look like. The database needs a master table where every metric has a standardised name, data type, and unit. For steps, the target schema looks like this:

- Target Column Name: *ik_steps*
- Data Type: Integer
- Time Aggregation: Daily Total (Midnight to Midnight, Local User Time)
- Metadata Required: *user_id*, *date*, *device_source_id*

2. Build the Mapping Dictionary The backend needs a configuration file (like a JSON dictionary or an SQL table) that explicitly tells the system which foreign keys belong to the Ikarians primary key. When imported dataset arrive at the Ikarians server, the system first checks the *device_source_id*. It then looks at this dictionary to know exactly which field to extract from that specific brand’s imported dataset.

3. Apply a Standard Time-Window Transformation Wearable brands send the data at different time aggregations or frequency, for example:

- Apple: Apple Health writes *stepCount* in tiny increments throughout the day.

The pipeline must standardise the time window. It should run a script that groups all entries by *user_id* and *date*, sums the step increments, and loads the single total integer into *ik_steps*.

4. The 3-Tier Data Pipeline Architecture To ensure the physician’s dashboard never crashes or displays wrong data, the database should be structured in three layers:

1. **Extract:** This is the step where data is pulled from the Apple/Garmin/Google APIs and saved in its raw form. This ensures we always have the original data for audit purposes or if we need to re-run algorithms.
2. **Transform:** This is where the script runs. The database acts as the normalization engine by matching the messy raw data against our data mapping dictionary and transforming it into the clean Ikarians standard variable.
3. **Load:** This is the final, clean table containing the output of Ikarians standard variable that the physician’s dashboard shows.

B. For the Subjective Data:

Based on the Data Source Catalogue compiled from the Apple, Samsung, Google, and Garmin documentation, the wearable APIs does leave gaps in the subjective, psychosocial, and qualitative pillars of the 6 longevity pillars. To address this, I would adapt using PROMIS (Patient-Reported Outcomes Measurement Information System) Short Forms to capture context that wearables can't currently detect.

All mentioned forms and screening tools are made available in the attached “Supplementary Document - Forms” file.

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